ZIP models

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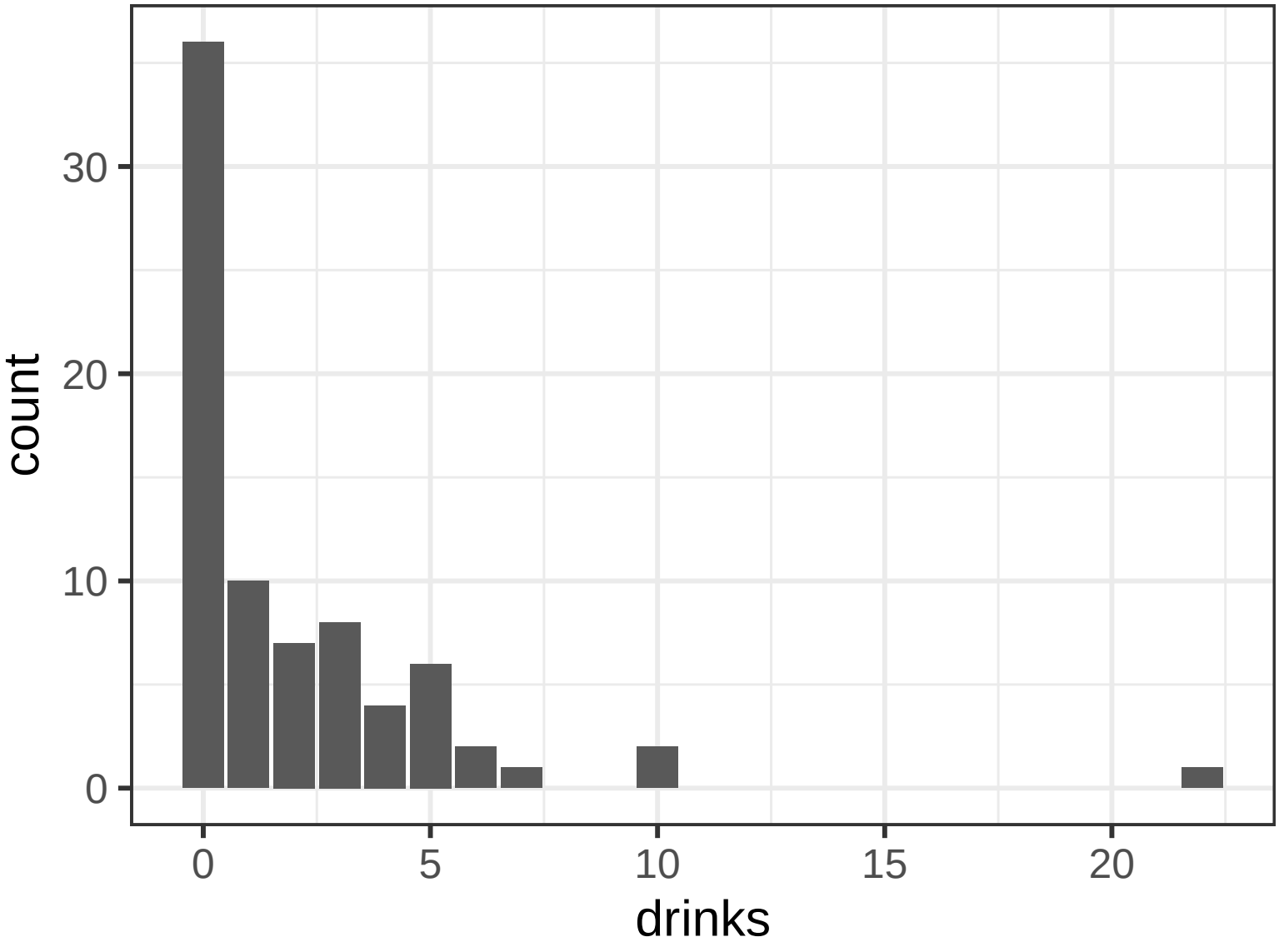
Last time: weekend drinking

Survey data from 77 college students on a dry campus (i.e., alcohol is prohibited) in the US. Survey asks students "How many alcoholic drinks did you consume last weekend?"

- `drinks`: number of drinks the student reports consuming
- `sex`: whether the student identifies as male
- `OffCampus`: whether the student lives off campus
- `FirstYear`: whether the student is a first-year student

Our goal: model the number of drinks students report consuming.

EDA: drinks



Zero-inflated Poisson (ZIP) model

In R

```
library(pscl)

m1 <- zeroinfl(drinks ~ sex + FirstYear + OffCampus,
               dist = "poisson", data = wdrinks)

m1$coefficients
```

```
## $count
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##     0.8010435     0.9834693    -0.1619318     0.3723520
##
## $zero
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##   -0.39618515   -0.07079078    0.89197114   -1.69137146
```

Activity

```
m1$coefficients
```

```
## $count
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##     0.8010435     0.9834693    -0.1619318     0.3723520
##
## $zero
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##   -0.39618515   -0.07079078    0.89197114   -1.69137146
```

What is the estimated probability that a female first-year student who lives on campus never drinks?

Activity

```
m1$coefficients
```

```
## $count
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##     0.8010435     0.9834693    -0.1619318     0.3723520
##
## $zero
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##   -0.39618515   -0.07079078    0.89197114   -1.69137146
```

What is the estimated average number of drinks for a second-year male student who lives off campus and sometimes drinks?

Activity

```
m1$coefficients
```

```
## $count
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##     0.8010435     0.9834693    -0.1619318     0.3723520
##
## $zero
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##   -0.39618515   -0.07079078    0.89197114   -1.69137146
```

What is the estimated probability that a male first-year student who lives off campus had at least one drink last weekend?

Activity: fitting ZIP models

Work on the class activity (course website).

Activity

Work on the class activity (handout).

Activity

- `cigsPerDay`: The number of cigarettes smoked per day during the study period.
- `education`: 1 = High School, 2 = Some College, 3 = College Degree, 4 = Advanced Degree.
- `male`: 1 = Male, 0 = Female.
- `age`: The age of the individual in years.
- `diabetes`: 1 if the individual has diabetes, 0 otherwise.

Why might a ZIP model be appropriate for this data? What would the latent variable represent?

Activity

	Estimate	Std. Error	z value	Pr(>|z|)	
(Intercept)	2.9585841	0.0350454	84.422	<2e-16	***
male	0.4621809	0.0109838	42.078	<2e-16	***
age	-0.0067425	0.0006782	-9.941	<2e-16	***
education2	0.0131422	0.0125969	1.043	0.297	
education3	0.0128553	0.0160200	0.802	0.422	
education4	-0.0187548	0.0171957	-1.091	0.275	
diabetes	-0.0166835	0.0393369	-0.424	0.671	

How would I interpret the estimated coefficient 0.462?